

ASTRA: Adaptive Satellite Task and Resource Allocator for LEO Earth Observation

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Abstract—Small Earth-observation satellites capture multi-spectral and Synthetic Aperture Radar data under intermittent contact and constrained onboard resources. We introduce ASTRA (Adaptive Satellite Task and Resource Allocator), a research prototype for the IASTAM 6.0 “Process or Transmit?” problem. We compare transmit-all, greedy-edge, a threshold heuristic, and an initial MaskablePPO policy in a seeded one-second-step simulator over five paired four-orbit episodes. The heuristic achieved the largest delivered-utility ratio (0.0286); greedy-edge completed the largest fraction of generated payloads (5.17%). The single-seed PPO checkpoint completed no tasks, exposing a limitation in current training rather than demonstrating RL inferiority. The simulator’s simplified thermal model can exceed its 50°C throttle threshold, and results are not flight-validated. Lyapunov, oracle, multi-seed learning, and hardware-calibration studies remain future work.

Index Terms—orbital edge computing, onboard task scheduling, energy- and thermal-aware computing, Lyapunov optimization, reinforcement learning, Low Earth Orbit, fault tolerance.

I. INTRODUCTION

The 2026 edition of the IEEE IAS Tunisia Annual Meeting (IASTAM 6.0) Technical Challenge, organized in collaboration with the Tunisian Space Association (TUNSA), addresses an operational bottleneck in Earth-observation data handling. Orbital edge computing has drawn attention as a way to move selected processing closer to sensors [1], [8]. Google’s Project Suncatcher is a proposed space-based AI infrastructure concept, not an operational orbital testbed [8].

Within this domain, Problem 1 (“Process or Transmit?”) targets an acute bottleneck in Earth Observation (EO): small satellites continuously capture high-resolution optical and Synthetic Aperture Radar (SAR) payloads (generating multi-megabyte payloads every 15 s), yet operate under three binding physical constraints:

- 1) **Battery Floor:** The simulator models a 100 Wh battery and clamps its state at a 30% minimum depth-of-discharge floor.
- 2) **Thermal Constraint:** Vacuum requires radiative heat rejection. The current simulator uses a first-order thermal proxy and pauses compute at 50°C; it does not enforce a hard cap, and the measured trace reaches 52.48°C.

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- 3) **Severe Downlink Bandwidth Choke:** Line-of-sight contact with ground stations is limited to short windows (120–180 s per 90-minute orbit, about 2.8% mean contact duty cycle) with an X-band rate of 20.0 Mbps (2.5 MB/s).

For every captured payload, the onboard system must choose among four actions: `PROCESS_NOW` (compression vs. inference), `STORE_QUEUE` (time-shifted execution in Mass Memory Unit, MMU), `TRANSMIT_RAW` (downlink during line-of-sight passes), or `DISCARD_DROP` (pruning low-utility or cloud-occluded data). The optimization objective is to maximize cumulative scientific and operational utility, which decays exponentially over latency.

Contributions:

- We implement paired benchmarks for transmit-all, greedy-edge, the threshold heuristic, and an initial MaskablePPO policy.
- We report eight outcome and resource metrics over five four-orbit evaluation seeds, including per-policy uncertainty.
- We identify an observed temperature excursion above the code’s 50°C throttling threshold, a limitation that motivates thermal-model and safety-guard validation.

II. RELATED WORK

A. Orbital and Edge Computing Systems

Orbital edge computing surveys describe the systems-level case for in-space processing [1]. Trabant is a recent serverless architecture proposal [2]; the Tiansuan constellation provides a cloud-native satellite case study [3]. Broader treatments of Space AI [4] and Earth-observation image processing [5] provide context on onboard classification and inference. FPGA accelerator surveys [6] and work on when to compute in space [7] inform the modeled trade-offs; the simulator’s numerical parameters remain assumptions rather than transferred empirical measurements.

B. Control and Scheduling Paradigms

Resource-constrained orbital scheduling is addressed via two dominant paradigms:

- 1) **Lyapunov Optimization & Drift-Plus-Penalty:** Prior work [10] motivates online scheduling; implementing and evaluating an orbital scheduler remains planned work.

- 2) **Constrained Reinforcement Learning (CMDP):** We train an initial MaskablePPO policy using the repository’s Gymnasium environment and invalid-action masks; this one-seed result is exploratory.

C. Added Value of This Work

Prior work motivates orbital edge processing and constrained scheduling. This interim study provides a reproducible comparison of two operational reference policies, the threshold heuristic, and one exploratory learned policy, and identifies model and learning gaps.

III. METHODOLOGY

A. Physical Models & Orbital Mechanics

The satellite orbital environment and hardware state at time t are formalized in Table I.

TABLE I
PRIMARY SYSTEM AND STATE VARIABLES

Symbol	Physical Definition	Nominal Value / Range
$T_{\text{orbit}}, T_{\text{sun}}$	Orbit period / Sunlit duration	5400 s / 3300 s (90 / 55 min)
$P_{\text{harvest}}(t)$	Solar power harvested	0.0 W (ecl) to 30.0 W peak
$\text{SoC}(t)$	Battery State-of-Charge	Floor ≥ 0.30 , Cap: 100.0 Wh
$T_{\text{chip}}(t)$	Processor temperature	Ambient 0°C , Ceiling $\leq 50.0^\circ\text{C}$
$G(t)$	Ground station contact flag	$\{0, 1\}$, 120–180 s pass (about 2.8% mean duty)
B_{downlink}	RF communication speed	20.0 Mbps = 2.5 MB/s (15.0 W)
P_{seu}	SEU bit-flip rate	10^{-5} s^{-1} cosmic radiation
$M_{\text{mmu}}, M_{\text{ram}}$	Mass Memory / RAM	32 000 MB / 4 000 MB

1) **Power and Battery Dynamics:** The simulator uses a 55-minute sunlight interval followed by a 35-minute eclipse. In sunlight, harvested power follows a sine-shaped profile from a 22.5 W base to a 30 W peak, with 30 s cosine tapers at sunlight boundaries; eclipse harvesting is zero.

Battery energy is updated by

$$E_{\text{batt}}(t+1) = \min(E_{\text{cap}}, E_{\text{batt}}(t) + [P_{\text{harvest}}(t) - P_{\text{draw}}(t)]\Delta t), \quad (1)$$

where $P_{\text{draw}} = P_{\text{base}} + P_{\text{compute}} + P_{\text{comm}}$, $P_{\text{base}} = 5 \text{ W}$, and radio draw is 15 W during downlink. The implementation has no charge/discharge efficiency; it clamps battery energy to the 30% floor on brownout and exits safe mode above 1.05 times that floor.

2) **Lumped Thermal Proxy:** The simulator uses a first-order thermal proxy rather than a Stefan–Boltzmann radiation model:

$$T_{\text{chip}}(t+1) = T_{\text{chip}}(t) + 0.15P_{\text{draw}}(t) - 0.10(T_{\text{chip}}(t) - T_{\text{amb}}). \quad (2)$$

This one-second update uses watts of total bus, compute, and communication draw, with $T_{\text{amb}} = 0^\circ\text{C}$. Compute pauses at or above 50°C , but this threshold is not a hard cap; the seed-42 two-orbit trajectory peaks at 52.48°C .

B. Workload, Queueing, and Utility Formulation

Payloads k arrive periodically every $\Delta t_{\text{gen}} = 15 \text{ s}$ across dual modalities:

- **Optical Multispectral (70%):** Size $S_k \in [5, 15] \text{ MB}$, initial utility $U_{0,k} \in [10, 100]$.

- **Synthetic Aperture Radar (30%):** Size $S_k \in [100, 300] \text{ MB}$, initial utility $U_{0,k} \in [10, 100]$.

Payload utility decays exponentially over latency ($t - t_{\text{capture}}$):

$$U_k(t) = U_{0,k} \cdot \exp(-\alpha_k(t - t_{\text{capture}})) \quad (3)$$

with decay parameter $\alpha_k \in [10^{-4}, 10^{-3}] \text{ s}^{-1}$.

Compute actions transform payload properties:

- **Compression:** Duration 2 s, $P_{\text{comp}} = 10 \text{ W}$, size factor $C_{\text{ratio}} = 0.30$, utility retention $\rho_{\text{val}} = 0.90$.
- **Inference:** Duration 5 s (with 2 s wake penalty), $P_{\text{inf}} = 20 \text{ W}$ (30 W wake), size factor $C_{\text{ratio}} = 0.001$, utility retention $\rho_{\text{val}} = 0.50$.

C. Decision Engine & Solution Schedulers

For each payload k , the agent selects an action $a_k \in \mathcal{A}$, where:

$$\mathcal{A} = \{\text{PROCESS_COMPRESS}, \text{PROCESS_INFER}, \text{DOWNLINK_RAW}, \text{STORE_QUEUE}, \text{DISCARD_DROP}\} \quad (4)$$

- 1) **Transmit-all:** Sends the oldest raw MMU payload during contact, without onboard processing or voluntary drops.
- 2) **Greedy-edge:** Processes the oldest eligible payload when queue capacity permits, then downlinks processed outputs during contact.
- 3) **Threshold heuristic:** Uses contact-aware downlink, SoC/temperature-gated processing, and stale-data eviction under storage pressure.
- 4) **MaskablePPO:** Uses normalized telemetry and top- K payload metadata with invalid-action masks. This initial learned policy is exploratory, not tuned or safety-certified.
- 5) **Planned methods:** Lyapunov scheduling and a non-causal MILP oracle are not implemented.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup and Results

Four policies share the same simulator configuration, four-orbit horizon (21,600 one-second steps), and five paired evaluation seeds (1001–1005); the heuristic is the reference. MaskablePPO was trained with seed 7 for 100,000 requested timesteps, with evaluation seeds disjoint from training. Table II reports episode means, sample standard deviations, and two-sided 95% Student- t intervals ($n = 5$). Decision quality divides delivered utility by the snapshot sum of generated payload base values. Memory use is time-averaged occupancy; energy includes modeled base, compute, and communication draw; latency averages completed deliveries only. Since only one PPO training seed was evaluated, its result does not measure training-seed variability.

The first PPO checkpoint completed no payload in the five held-out episodes. Its lower energy is therefore an inactivity outcome, not a demonstrated energy-efficiency gain; this initial

TABLE II
PAIRED POLICY COMPARISON ACROSS FIVE FOUR-ORBIT EPISODES (MEAN; SAMPLE SD; 95% CI)

Metric	Transmit-all	Greedy-edge	Heuristic	MaskablePPO (seed 7)
Decision quality	.00183; .00115; [.00040,.00326]	.00470; .00105; [.00339,.00601]	.02860; .00142; [.02683,.03036]	0; 0; [0,0]
Completed-task rate	.01569; .00212; [.01307,.01832]	.05167; .00805; [.04168,.06166]	.01597; .00049; [.01536,.01658]	0; 0; [0,0]
Energy (kJ)	117.150; .340; [116.728,117.571]	290.882; 1.761; [288.695,293.068]	290.882; 1.761; [288.695,293.068]	206.012; 4.782; [200.075,211.949]
Delivery latency (s)	9658.52;1995.69;[7180.95,12136.10]	9922.94;1289.76;[8321.74,11524.14]	753.41;121.36;[602.75,904.07]	N/A (no delivery)
MMU utilization (%)	82.81;1.96;[80.39,85.24]	38.16;1.91;[35.79,40.54]	37.96;1.90;[35.60,40.32]	48.77;3.63;[44.27,53.28]
RAM utilization (%)	0.0;[0,0]	.173;.003;[.169,.177]	.173;.003;[.169,.177]	.439;.036;[.394,.484]
SEU events / episode	0.0;[0,0]	0.0;[0,0]	0.0;[0,0]	0.0;[0,0]
Safe-mode events / episode	0.0;[0,0]	0.0;[0,0]	0.0;[0,0]	0.0;[0,0]

run should motivate reward and training diagnostics. No SEUs occurred in these five evaluation episodes, which is too few to assess rare-event recovery. Across a separate two-orbit seed-42 trace, SoC ranged from 92.37% to 100%, while temperature ranged from 0.75°C to 52.48°C. The 50°C threshold throttles processing but does not bound simulated temperature.

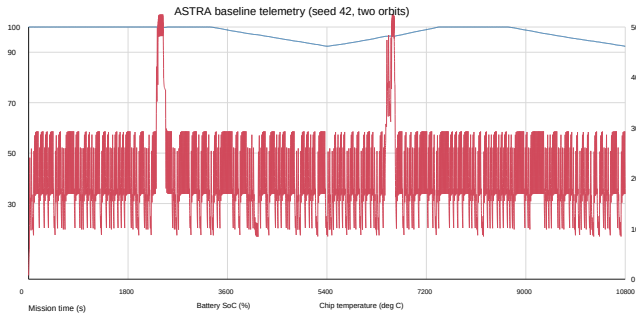


Fig. 1. Heuristic trajectory over two orbits (seed 42). Temperature exceeds the 50°C throttle threshold.

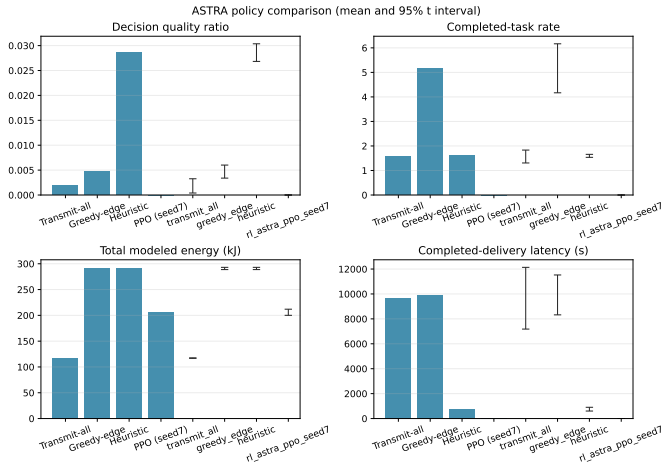


Fig. 2. Policy comparison; bars show means and error bars show 95% t intervals. PPO completes no tasks in this run.

Greedy-edge’s larger completion fraction does not imply higher delivered utility: the utility ratio is lower than for the threshold heuristic. The paired results are preliminary ($n = 5$), and the PPO result uses one training seed. Transmit-all and greedy-edge are operational reference policies, not optimized

implementations. Lyapunov scheduling and the MILP oracle remain future work.

B. Phase 3 Validation Plan

Next, diagnose sparse-reward learning and improve PPO training using multiple independent training seeds and validation episodes, then evaluate a selected checkpoint on a fresh test set. Further work includes Lyapunov scheduling, a non-causal MILP oracle, and stress tests for SEU bursts, battery degradation, and missed contacts.

V. CONCLUSION

We implemented a paired comparison of transmit-all, greedy-edge, the threshold heuristic, and an initial MaskablePPO policy in one four-orbit simulator benchmark. The heuristic achieved the highest delivered-utility ratio (0.0286), while greedy-edge completed more tasks (5.17%). The single-seed PPO policy delivered no tasks and needs training/reward redesign before it can support a useful comparison. The simplified thermal model can exceed its throttle threshold; all results are synthetic, not flight-validated. Follow-up work should address these limits and add multiple PPO seeds, Lyapunov scheduling, and an oracle benchmark.

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